From No Clock to Relational Time:
Page–Wooters Genesis, AR(∞)→AR(2), Clock Births, Multiplicities, and a Finite-Loop GR Bridge

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Abstract

We reformulate the program "from no intrinsic clock to emergent clocks, multiplicities, and geometry" in a relational (Page–Wooters) setting. Layer 0 is a *clockless label space* on which regulators and probes act; temporal order appears only after a gauge choice. The conditioned dynamics of residue streams obey an exact decimated law derived from a 2×2 transfer engine. We show how an AR(∞) parent with one dominant conjugate pair admits a controlled AR(2) surrogate, and how decoherence arises as phase diffusion quantified by $D = \frac{1}{2}S_\delta(0)$ under demodulation. Clock *births* are produced by net-zero probes with closed-form (finite-ring) Green functions; a binary PBC/APBC probe opens/closes a clock-sector gap and flips an *operational* trace estimator $\widehat{V}_L$ while the intrinsic $V_L = \mathrm{tr}(M^L)$ remains invariant. Finally, we provide a finite-loop holonomy–Levi–Civita bridge with explicit units and a clear error budget, extend to directed graphs, and include code-ready estimators.

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1 Layer 0: No intrinsic time — labels, regulators, and probes

At the foundational layer we assume only a set of labels $\mathcal{E}$ (no metric, order, or topology). *Regulators* are symmetry operations $R \in \mathcal{R}$ acting on functions $f : \mathcal{E} \to \mathbb{C}^2$ (e.g., group representations, quotientings), while *probes* are linear functionals extracting finite summaries. No sequential structure is postulated. A *gauge choice* is a section that identifies a discrete cover of a regulator orbit with $\mathbb{Z}$, yielding an index map $k \mapsto e_k$ (bookkeeping). Only after this choice do recursions or boundary conditions appear; different gauges are physically equivalent if they produce indistinguishable probe statistics.

Resolution of the "No Clock" paradox. Objects such as Möbius twists, difference operators, or Dirac boundary conditions *do not* assert pre-existing time; they are *regulators* and *diagnostics* on label classes. Temporal order emerges when a gauge is *fixed* and validated by acceptance tests (e.g., stability of inferred invariants across gauges).

2 Relational conditioning (Page–Wooters) in discrete form

Let the composite "clock $\otimes$ system" be constrained by a discrete PW equation $\widehat{\mathcal{C}}\,|\Psi\rangle = 0$ with $\widehat{\mathcal{C}}$ block-linear in a shift on the clock labels and a transfer on the system. Conditioning on a clock outcome (a probe event) and choosing a gauge section $\mathbb{Z} \to \mathcal{E}$ yields linear residue dynamics on the system observables $\{x_k\}$ with a *transfer engine* described next.

3 Engine: $\mathrm{GL}(2, \mathbb{C})$ transfer, exact decimation, and speed bound

Transfer engine and invariants

A one-step transfer $M \in \mathrm{GL}(2, \mathbb{C})$ (gain/loss allowed). Define

$$V_1 = \operatorname{tr} M, \qquad Q = \det M, \qquad V_\ell = \operatorname{tr}(M^\ell), \qquad Q_\ell = (\det M)^\ell.$$

For a scalar observable $\{x_k\}$, decompose into residue tracks $y_t^{(r)} = x_{r+\ell t}$.

Exact decimated two-step law (Cayley–Hamilton)

For each residue r and stride ℓ,

$$y_{t+2}^{(r)} - V_\ell\, y_{t+1}^{(r)} + Q_\ell\, y_t^{(r)} = 0, \qquad r_\pm(\ell) = \frac{V_\ell \pm \sqrt{V_\ell^2 - 4Q_\ell}}{2}. \tag{1}$$

Sketch

Cayley–Hamilton on M^ℓ: $M^{2\ell} - V_\ell M^\ell + Q^\ell \mathsf{I} = 0$. Apply to state Ψ_k and project to any linear observable.

Unitary vs transfer engine. If $M \in \mathrm{SU}(2)$ then $Q = 1$ and $|r_\pm| = 1$. For general $M \in \mathrm{GL}(2, \mathbb{C})$, $Q \neq 1$ captures amplification/attenuation; one may optionally normalize $\widetilde{M} := M/\sqrt{\det M} \in \mathrm{SL}(2, \mathbb{C})$ for trace identities.

Envelope and speed limit (with units). Introduce a locality envelope

$$\mathsf{E} := \exp(-\sigma^2 \kappa\, \tau\, \mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha), \qquad s := \sigma^2 \kappa\, \tau \ \ \text{(dimensionless)}, \tag{2}$$

with L_α a (possibly PSD) graph Laplacian, $\Lambda_{\max}$ its spectral radius, and τ the step. Then

$$v_{\max} \lesssim \frac{a \sup_{\mathbf{q}} \|\nabla_{\mathbf{q}} \varepsilon(\mathbf{q})\|}{1 + s\,\Lambda_{\max}^2} \ \ \text{(Fourier band phase } \varepsilon\text{)}, \qquad v_{\max} \lesssim \frac{2a\,\|\widehat{K}\|\,\Delta}{1 + s\,\Lambda_{\max}^2} \ \ \text{(graph)}. \tag{3}$$

4 AR(∞) parent $\to$ AR(2) surrogate and consistency regime

Assume a coarse observable has an AR(∞) parent with one dominant conjugate pair $r_\pm = \rho e^{\pm i\theta}$ ($\rho \leq 1$) and a stable tail.

Padé [1/1] surrogate captures the pair exactly

The AR(2) surrogate

$$x_{t+2}^{(2)} - V\, x_{t+1}^{(2)} + Q\, x_t^{(2)} = u_t, \qquad V = 2\rho\cos\theta, \quad Q = \rho^2, \tag{4}$$

has characteristic roots $r_\pm = \rho e^{\pm i\theta}$; the tail folds into an MA(∞) innovation u_t.

When is AR(2) adequate? Empirically require a dominance ratio

$$\frac{|r_+|}{\max_{j \geq 3} |p_j|} \gtrsim 1.2 \quad \text{and white AR(2) residuals (Ljung–Box at 5\%).}$$

Specialize to $Q = 1$ (unitary engine) by setting $\rho = 1$ and $V = 2\cos\theta$.

5 Superposition and decoherence as phase diffusion

Project onto the rotating eigenmode, demodulate by the fitted $\hat{r}_+$, and write the phasor recursion

$$a_{t+1} = \rho\, e^{i(\theta + \delta_t)}\, a_t, \qquad \mathbb{E}\,\delta_t = 0. \tag{5}$$

Then off-diagonal coherence decays as

$$|\mathcal{C}(t)| := \frac{|\mathbb{E}[a_t \overline{a_0}]|}{\mathbb{E}|a_0|^2} \approx \rho^t\, e^{-\frac{1}{2} S_\delta(0)\, t}, \qquad D := \tfrac{1}{2} S_\delta(0). \tag{6}$$

From additive noise to phase noise. If $a_{t+1} = r_+ a_t + \xi_t$ with small complex innovation, then $\delta_t \approx \mathrm{Im}(\xi_t/(r_+ a_t))$; estimate $S_\delta(0)$ from the demodulated, amplitude-normalized residual.

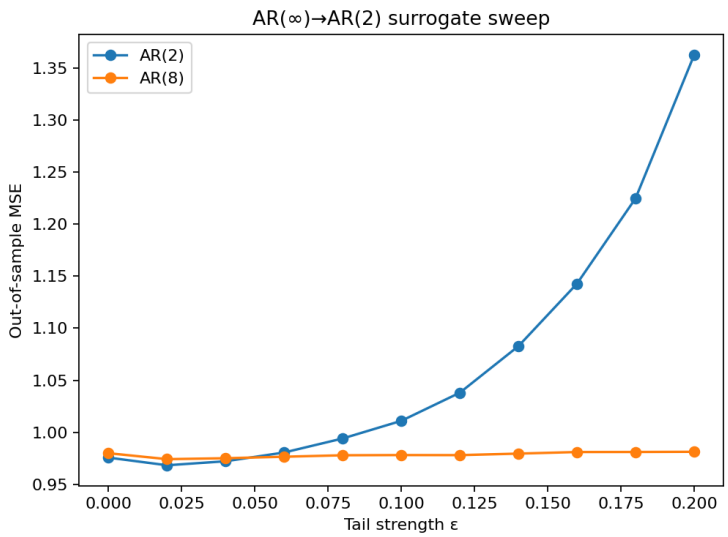

Figure 1: AR(∞)→AR(2) surrogate sweep: fit quality across tail strengths and dominance ratios.

One-pass estimator for (ρ, D) (pseudocode)

```
Fit AR(2): y_{t+2} = V y_{t+1} - Q y_t + u_t  ->  roots r_+/-, pick r_+ (arg in (0, pi))
Demodulate: z_t = y_t * (conj(r_+))^t
Phase increments: delta_t = arg(z_{t+1}*conj(z_t)) - arg(r_+)  (wrapped to (-pi, pi])
Estimate S_delta(0) via Bartlett at zero; return rho=|r_+|, D = 0.5*S_delta(0)
```

Robustness. Gate out very small $|a_t|$ or weight by $|a_t|^2$ when estimating $S_\delta(0)$ to stabilize phase-noise estimates at low amplitudes.

PBC/APBC and D. A PBC (Ramond) clock admits a zero mode; APBC (NS) opens a gap $\sim \pi/T$. This *modifies* the low-frequency phase-noise power $S_\delta(0)$ through clock–tail coupling (sign model-dependent); test empirically by the above estimator. See also Sec. 8.

6 Gravity as AR(∞) Back–Reaction of Clocks (Newtonian Limit)

From AR(∞) to a static slowness field. A coarse observable admits a two–pole + tail representation

$$x_{t+1} - Vx_t + Qx_{t-1} = \sum_{j\geq 1} b_j x_{t-j} + u_t, \qquad V = 2\rho\cos\theta, \quad Q = \rho^2.$$

Demodulate by the dominant pole $r_+ = \rho e^{i\theta}$: $z_t = x_t r_+^{-t}$. The tail induces phase diffusion $z_{t+1} = e^{i(\delta_t - \phi(x))} z_t$ with zero–mean δ_t and a slow, static $\phi(x)$ (phase–slowness field). Define the source density

$$\rho_{\text{tail}}(x) = \beta\, S_\delta(x, 0),$$

with $S_\delta(x, 0)$ the zero–frequency power of phase increments (estimated after AR(2) demodulation).

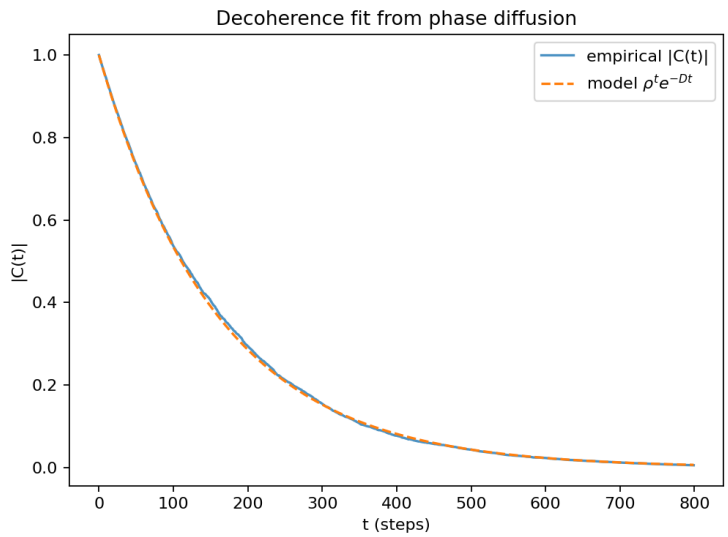

Figure 2: Decoherence fit $|\mathcal{C}(t)|$ vs the model $\rho^t e^{-Dt}$ with $D = \frac{1}{2} S_\delta(0)$.

Envelope as a gradient penalty. The envelope $E = \exp\Big(-s\, \mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha\Big)$ with $s = \sigma^2 \kappa\, \tau$ penalizes spatial phase gradients, yielding the static functional

$$\mathcal{J}[\phi] = \frac{\kappa_s}{2} \int |\nabla \phi|^2\, dV - \int \phi\, \rho_{\text{tail}}\, dV, \qquad \kappa_s := s\, c_{\text{env}}.$$

The Euler–Lagrange equation gives $-\kappa_s \nabla^2 \phi = \rho_{\text{tail}}$. With $\Phi := \alpha\phi$ and $4\pi G := \alpha^{-1} \kappa_s^{-1} \beta$, we obtain the Poisson equation

$$\boxed{\nabla^2 \Phi = 4\pi G\, \rho\ ,} \qquad \rho := \rho_{\text{tail}}.$$

Interpretation. Φ is the emergent Newtonian potential (phase–time dilation); ρ is the density of low–frequency tail power (or clock–birth intensity). In the pure AR(2) limit ($\delta_t \equiv 0$) we have $\rho = 0 \Rightarrow \nabla^2 \Phi = 0$. Hence gravity is an AR(∞) effect, not an AR(2) effect.

Consistency with the binary probe. The PBC/APBC dial modifies access to $\omega \approx 0$ in the clock sector (zero mode vs gap), changing $S_\delta(0)$ and thus ρ operationally, while the intrinsic system trace $V_\ell = \mathrm{tr}(M^\ell)$ is unchanged (the observed estimator $\widehat{V}_\ell$ flips sign after conditioning under APBC).

7 Clock births by net-zero probes and finite-ring Green functions

Inject a net-zero source s on a macrocycle of length L (e.g. $s_m = +\delta_{m,0} - \delta_{m,L/2}$ for even L). The AR(2) response x solves

$$x_{m+1} - V x_m + Q x_{m-1} = s_m, \qquad \sum_{m=0}^{L-1} s_m = 0. \tag{7}$$

Finite ring Green function (enforcing net-zero). On a ring,

$$G_n^{(L)} = \frac{1}{L}\sum_{k=0}^{L-1} \frac{e^{2\pi i k n/L}}{V - 2\sqrt{Q}\cos(2\pi k/L)}, \tag{8}$$

reducing to $G_n = \sin(n\phi)/\sin\phi$ for $Q = 1$, $V = 2\cos\phi$ as $L \to \infty$. *Net-zero source:* the DC Fourier bin ($k = 0$) is not excited, so there is no DC response. Handle the resonant case $V = 2\sqrt{Q}$ by the double-root limit/pseudoinverse.

The resulting response $x = G^{(L)} * s$ exhibits clean "births" of oscillatory residues shaped by the envelope.

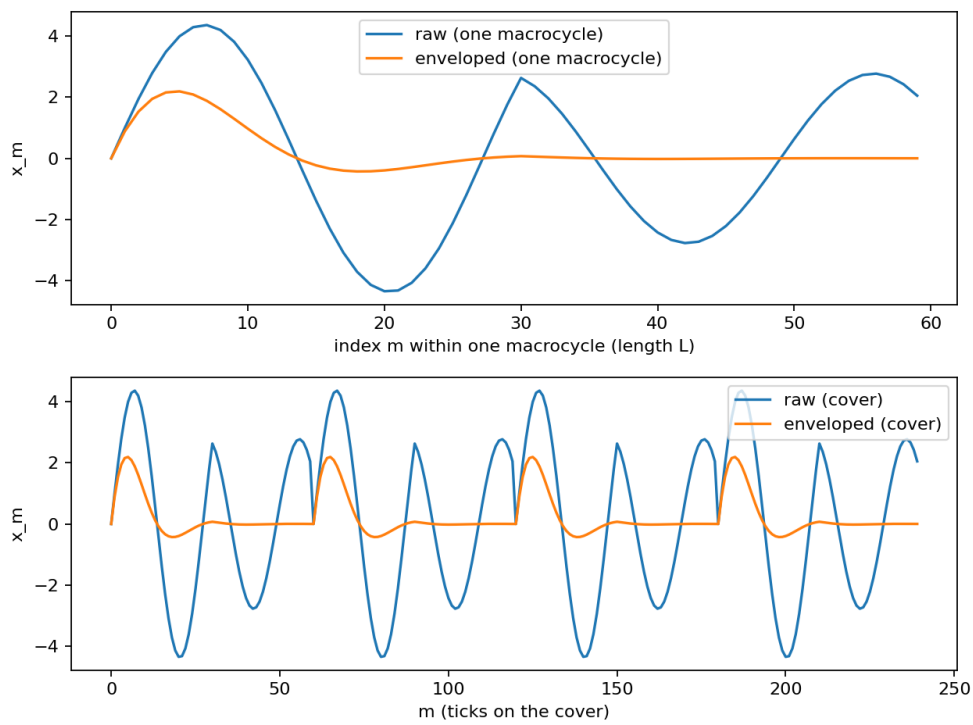

Figure 3: Clock births from a net-zero source on one macrocycle and repeated on the universal cover.

8 Binary probe: PBC zero mode vs APBC gap (operational flip)

A discrete Dirac on a time circle of period T has eigenfrequencies $\omega_n = \frac{2\pi}{T}(n + \frac{s}{2})$ for $s = 0$ (PBC) or $s = 1$ (APBC): PBC admits a zero mode; APBC has $|\omega| \geq \pi/T$ (gap). In our operational pipeline, twisting the *clock* by APBC injects a π-phase over one macrocycle in the conditioned residue at stride L, which flips the *estimator* $\widehat{V}_L$ derived from the data, while the intrinsic $V_L = \mathrm{tr}(M^L)$ is unchanged (a system invariant). This matches the spectral picture and the checksum demo reported in the supporting note. *Caption fix for plots:* "$\widehat{V}_L$ flips sign under APBC; V_L is invariant."

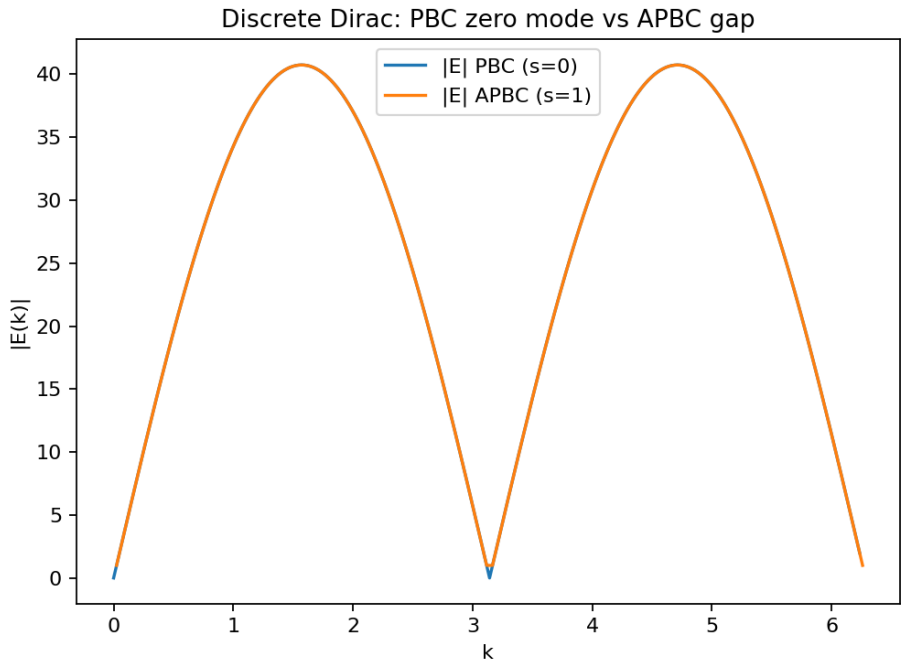

Figure 4: Discrete Dirac on a time circle: PBC shows a zero mode; APBC is gapped by $\sim \pi/T$.

Least-squares sign flip at stride L

The AR(2) LS estimator on the stride-L residue is $\widehat{V}_L = \dfrac{\sum_t (y_{t+2}\overline{y_{t+1}} + y_t\overline{y_{t+1}})}{\sum_t |y_{t+1}|^2}$. APBC sends $y_t \mapsto (-1)^t y_t$. The numerator gains a factor (-1) while the denominator is unchanged; hence $\widehat{V}_L \mapsto -\widehat{V}_L$ (and $\widehat{Q}_L$ is unchanged).

9 Multiplicity and species: operational labels and how to measure them

We label species by operational invariants

$$\Lambda = (\ell,\ V_\ell,\ Q_\ell,\ r_\pm = \rho e^{\pm i\theta},\ \chi \in \{\pm 1\},\ (c_0, c_1),\ \text{hol}),$$

where χ is the PBC/APBC parity measured at $L = \ell$, (c_0, c_1) are stream checksums (TKNN-like integers), and "hol" is the holonomy class (circle vs Möbius/Pin$^\pm$).

How to compute (c_0, c_1) quickly. Over a k-window, accumulate two linear invariants (real/imag parts) of $\sum_k d_k r_+^k$; tri-site local moves preserve them. A soft penalty with weight ν stabilizes reconstruction; normal equations $(I + \nu A^\top A)d = m + \nu A^\top b$ yield digits d (we reserve μ for the spectral gap of $L_\alpha^\dagger L_\alpha$).

Independence scale. If $L_\alpha^\dagger L_\alpha$ has spectral gap $\mu > 0$, then $\mathrm{Cov}(y^{(r)}, y^{(r+d)}) \sim e^{-\mu d}$. In practice take $d \gtrsim 3\,\mu^{-1}$ to treat residue tracks as independent samples.

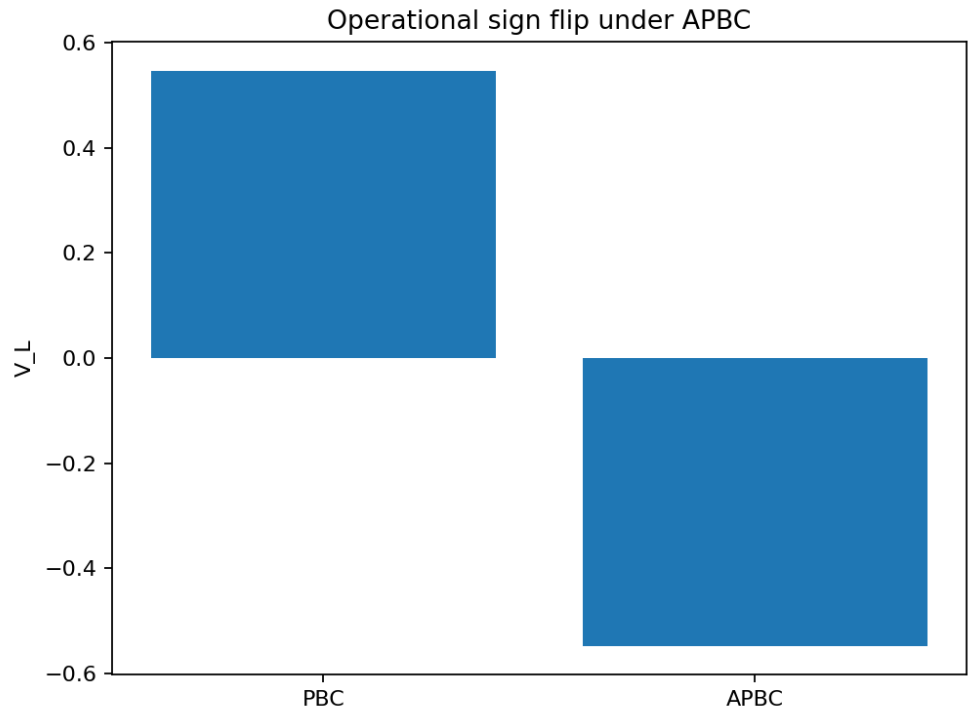

Figure 5: Operational test: $\widehat{V}_L$ flips sign under APBC while V_L remains invariant.

10 Finite-loop GR bridge with units, assumptions, and interpretation

Let phases φ^a push forward to a coframe $e^a = \Theta^a{}_i\, dx^i$; define $F^a{}_{ij} = \partial_i \Theta^a{}_j - \partial_j \Theta^a{}_i$, the Levi–Civita connection ω, and curvature Ω. Compare envelope holonomy $P_{\mathrm{env}}(C)$ with Levi–Civita $H_{\mathrm{LC}}(C)$ around loop C with spanning surface $\Sigma(C)$:

Finite-loop bridge (error budget with units)

$$\|P_{\mathrm{env}}(C) - H_{\mathrm{LC}}(C)\| \;\le\; C\Big(\underbrace{\sqrt{\int_{\Sigma(C)} \|F\|^2\, dA}}_{\text{antisym. defect}} + \underbrace{h}_{\text{mesh}} + \underbrace{\ell^3 \|\nabla\Omega\|_\infty}_{\text{curv. var.}} + \underbrace{\mu^{-1}\, s\, \ell}_{\text{envelope leakage}}\Big), \tag{9}$$

with $s = \sigma^2 \kappa\, \tau$, μ the spectral gap (units L^{-2}) of $\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha$, loop diameter ℓ, and mesh h. Norms are operator norms on $\mathfrak{so}(d)$; constants absorb unit conversions.

Idea of proof. (i) Strang splitting + BCH on edges (commutator budget $\propto s$); (ii) non-Abelian Stokes to replace edge-ordered exponentials by a surface integral (curvature budget); (iii) Cartan + Stokes + Poincaré/trace on rectangles, then polygonal refinement (defect + mesh budgets).

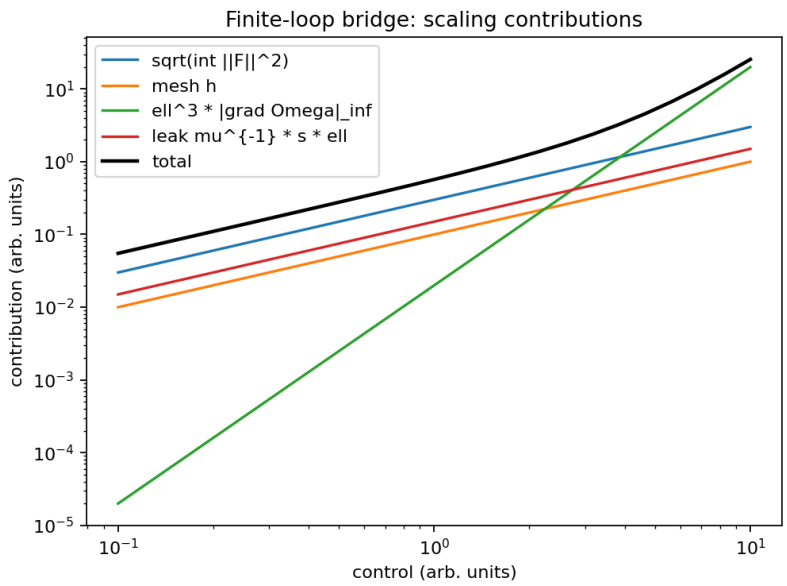

Figure 6: Finite-loop bridge calibration: scaling of $\|P_{\text{env}} - H_{\text{LC}}\|$ against the four error-budget terms.

11 Directed/weighted graphs (safe PSD Laplacian) and scaling knobs

For directed or weighted A, use the PSD surrogate

$$\mathsf{L}_{\text{psd}} = (\mathsf{I} - \alpha\mathsf{A})^{\dagger}(\mathsf{I} - \alpha\mathsf{A}), \qquad 0 < \alpha < \|\mathsf{A}\|_2^{-1},$$

and replace spectral radii by singular values throughout. The speed and bridge bounds retain their form with constants adjusted.

Knobs → effects (quick map).

Knob	Primary effect	Practical setting
$s = \sigma^2\kappa\,\tau$	locality ↑, $v_{\max}$ ↓	target $s\,\Lambda_{\max}^2 \simeq 1$ for 50% speed cut
γ (curl penalty)	$\|F\|$ suppression	raise until $\sqrt{\int \|F\|^2} \leq 0.01\,\text{Area}$
β (pushforward tie)	enforce $\Theta \approx \nabla\varphi$	set $\beta \approx \gamma a^2$
APBC twist	gap/zero-mode toggle	A/B test $\widehat{V}_L$, measure ΔD

12 Minimal experiments (falsifiable), figures, and captions

- **PBC vs APBC spectrum (clock sector):** show zero mode vs gap on the time circle (discrete Dirac); caption: "PBC has a zero mode; APBC opens a gap $\sim \pi/T$."
- **Operational flip:** plot $\widehat{V}_L$ from the conditioned residue under PBC vs APBC; caption: "$\widehat{V}_L$ flips sign under APBC; V_L is invariant."

- **AR(∞)→AR(2) adequacy:** sweep tails; report residual whiteness (LB test) and MSE vs dominance ratio.
- **Decoherence:** estimate (ρ, D) from demodulated phase; overlay $|\mathcal{C}(t)|$ vs $\rho^t e^{-Dt}$.
- **Bridge calibration:** vary (s, h, ℓ) on a grid with known Ω; verify slopes in (9).

13 Symbols and units (first-use table)

Symbol	Units	Meaning
a	length	lattice spacing
τ	time	step duration
$s = \sigma^2 \kappa \tau$	1	envelope strength (dimensionless)
$\Lambda_{\max}$	1	spectral radius of L_α (or sing. value)
Δ	1	max graph degree
$M \in \mathrm{GL}(2, \mathbb{C})$	—	transfer step; $V = \mathrm{tr} M$, $Q = \det M$
V_ℓ, Q_ℓ	—	stride-ℓ invariants, $\mathrm{tr}(M^\ell)$, $(\det M)^\ell$
$r_\pm = \rho e^{\pm i\theta}$	—	engine/AR roots
$\varepsilon(\mathbf{q})$	rad/step	band phase (Fourier)
$\Theta, F, \omega, \Omega$	mixed	coframe, antisym. defect, connection, curvature
h, ℓ	length	mesh size, loop diameter
μ	L^{-2}	spectral gap of $\mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha$
D	1/step	dephasing rate ($\frac{1}{2} S_\delta(0)$)

A Proof snippets and derivations

A. Cayley–Hamilton ⇒ decimated law

Apply $\chi_\ell(M^\ell) = 0$ to M^ℓ; project as in (1).

B. Finite-ring $G^{(L)}$ under net-zero forcing

Fourier-diagonalize the ring operator to obtain (8). Under a net-zero source the $k = 0$ (DC) bin is absent (no DC response); handle the $V = 2\sqrt{Q}$ resonance by the double-root limit/pseudoinverse.

C. Strang+BCH and leakage term

Write the edge step $U_e = \mathsf{E}^{1/2} T \mathsf{E}^{1/2}$ with $T = e^{\tau \widehat{K}}$; BCH yields nested-commutator norms $\propto s$; concatenation over a loop plus the gap μ controls leakage as $\mu^{-1} s\, \ell$.

B Reproducibility: compact code cells (language-agnostic)

AR(2) fit and dephasing estimator

```
# Given complex residue y[0..N-1]
# Fit AR(2): y_{t+2} = V y_{t+1} - Q y_t + u_t
Vhat,Qhat = argmin_VQ sum_t |y[t+2]-V*y[t+1]+Q*y[t]|^2
rplus = 0.5*(Vhat + sqrt(Vhat**2 - 4*Qhat))
```

```
# Demodulate and phase increments
z[t] = y[t] * conj(rplus)**t
delta[t] = wrap(angle(z[t+1]*conj(z[t])) - angle(rplus))
# Bartlett-at-zero for S_delta(0); D = 0.5*S_delta(0)
```

C Notes on operational vs intrinsic claims

Intrinsic quantities (e.g. $V_L = \mathrm{tr}(M^L)$) belong to the system and are invariant under clock twists; *operational* estimators (e.g. $\widehat{V}_L$ from conditioned residues) *can* change under PBC/APBC as the probe modifies the clock sector. Figures and statements are labeled accordingly.